

5W STEREO AUDIO AMPLIFIER

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Here is how we can build a 5W stereo audio power amplifier based on TEA 2025 IC and two potentiometers for left and right volume controls. Its power output is 2.5W on left and right channels each with a 12V DC power supply. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the 5W stereo audio amplifier is shown in Fig. 2. It is built around TEA 2025 (IC1), two 10-kilo-ohm potentiometers (VR1 and VR2), two 8-ohm, 2.5W loudspeakers (LS1 and LS2) and a few other components.

TEA 2025 is a 16-pin IC, whose pin 16 is connected to positive terminal of +12V supply. Electrolytic capacitor C4 is connected between +12V and ground. Pins 4, 5, 9, 12 and 13 of IC1 are grounded.

Left audio input (left in) and right audio input (right in) are connected

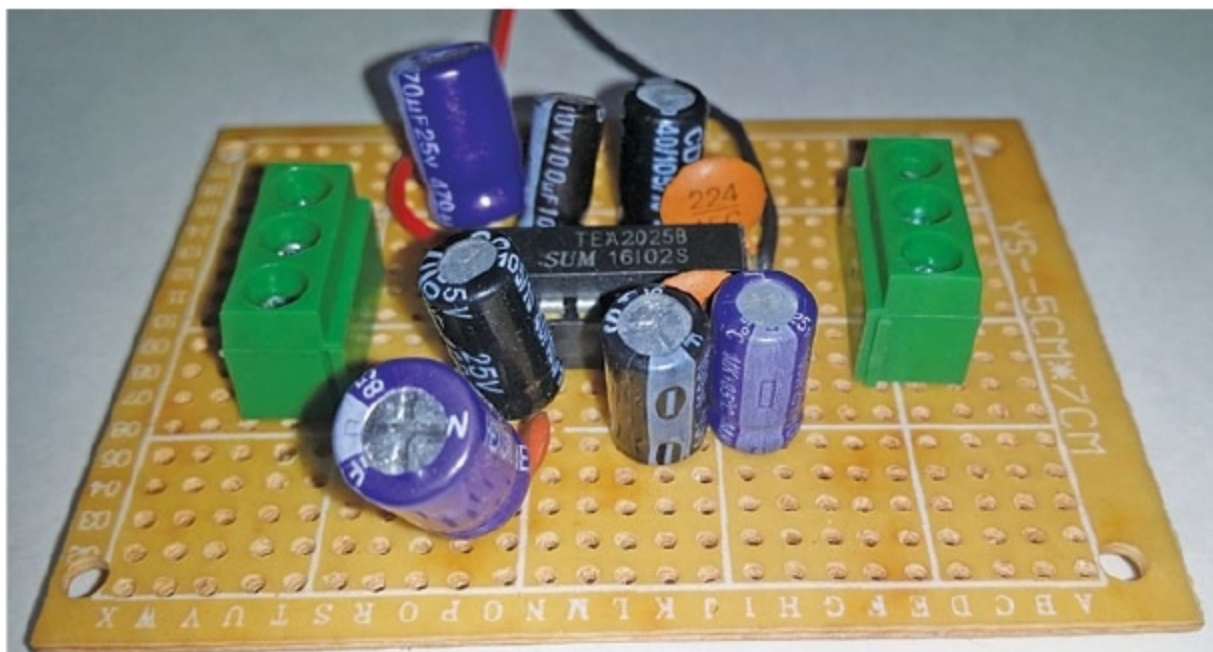


Fig. 1: Author's prototype

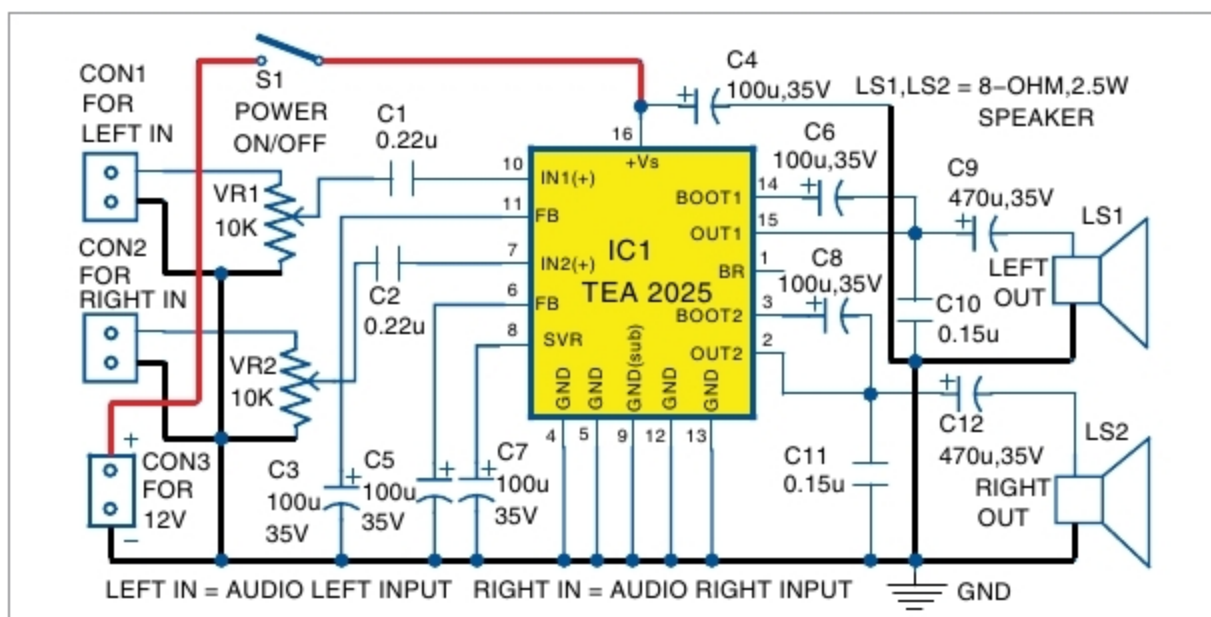


Fig. 2: Circuit diagram of 5W stereo audio amplifier

across connectors CON1 and CON2, respectively. Left and right audio inputs are further given to inputs of TEA 2025 via VR1 and VR2, and C1 and C2, respectively. Audio is amplified by IC1, and its left and right audio

outputs are available across LS1 and LS2, respectively.

Construction and testing

An actual-size PCB layout of the 5W stereo audio amplifier is shown in

PARTS LIST

Semiconductors:

IC1 - TEA 2025 stereo audio amplifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

VR1, VR2 - 10-kilo-ohm potentiometer

Capacitors:

C1, C2 - 0.22 μ F ceramic disk

C3-C8 - 100 μ F, 35V electrolytic

C9, C12 - 470 μ F, 35V electrolytic

C10, C11 - 0.15 μ F ceramic disk

Miscellaneous:

CON1, CON2 - 2-pin connector

CON3 - 2-pin terminal connector

S1 - On/off switch

LS1, LS2 - 8-ohm, 2.5W loudspeaker

- 16-pin IC socket/base

- 12, 1A power supply

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